Wasiur R. KhudaBukhsh

List of publications

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In mathematics, authors are conventionally listed alphabetically. Links to all papers are available at www.wasiur.xyz/Publications.

Preprints and submitted papers

1. **Wasiur R. KhudaBukhsh** and Yangrui Xiang. Cutoff profiles and boundary effects in the togashi–kaneko model of autocatalytic reactions with weak mutations. 2026. Submitted.
2. Kushankur Dutta, Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, and Grzegorz A. Rempała. Coloured epidemic models: functional law of large numbers and propagation of chaos. 2026. Submitted.
3. Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, M. Gabriela M. Gomes, and Grzegorz A. Rempała. From individual-based stochastic epidemics to heterogeneous SIR equations. 2026. Submitted.
4. Yordan P. Raykov, Hengrui Luo, Justin D. Strait, and **Wasiur R. KhudaBukhsh**. Shrinkage priors for Bayesian substitute confounders. 2026.
5. Arnab Ganguly and **Wasiur R. KhudaBukhsh**. Fluctuation analysis of the standard QSSA for a stochastic multistage Michaelis–Menten model. 2026. Submitted.
6. Olga Izyumtseva and **Wasiur R. KhudaBukhsh**. Self-intersection local times for Volterra Gaussian processes in stochastic flows with interaction. 2026. Submitted.
7. J. D. Harborne, **Wasiur R. KhudaBukhsh**, and John R. King. Deterministic and stochastic asymptotics for a multiscale epidemic model. 2026. Revision requested.
8. Yordan P. Raykov, Hengrui Luo, Justin D. Strait, and **Wasiur R. KhudaBukhsh**. Kernel-based estimators for functional causal effects. 2026.
9. **Wasiur R. KhudaBukhsh** and Yangrui Xiang. Mixing time for a noisy SIS model on graphs. 2026. Submitted.
10. Karim S. Elsayed, Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, and Amr Rizk. Stochastic analysis of entanglement-assisted quantum communication channels. 2025. Revision requested.
11. Olga Izyumtseva and **Wasiur R. KhudaBukhsh**. Local times of self-intersection and sample path properties of Volterra Gaussian processes. 2025. Revision requested.

Peer-reviewed journal articles

12. Arnab Ganguly and **Wasiur R. KhudaBukhsh**. Statistical inference for a multiscale stochastic model of enzyme kinetics via propagation of chaos. *Stochastic Processes and their Applications*, 2027. Accepted for publication.
13. Arnab Ganguly and **Wasiur R. KhudaBukhsh**. Asymptotic analysis of the total quasi-steady state approximation for the Michaelis–Menten enzyme kinetic reactions. *Journal of Mathematical Analysis and Applications*, 561(1):130551, 2026.
14. Yi Fu, Hye-Won Kang, **Wasiur R. KhudaBukhsh**, Lea Popovic, Grzegorz A. Rempała, and Ruth J. Williams. Fragility in a Togashi–Kaneko stochastic model with mutations. *SIAM Journal on Life Sciences*, 1(2):202–228, 2026.

15. Riccardo Corradin, Luca Danese, **Wasiur R. KhudaBukhsh**, and Andrea Ongaro. Model-based clustering of time-dependent observations with common structural changes. *Statistics and Computing*, 36(7), 2026.
16. Alexander E. Zarebski, Nefel Tellioglu, Jessica E. Stockdale, Julie A. Spencer, **Wasiur R. KhudaBukhsh**, Joel C. Miller, and Cameron Zachreson. Including frameworks of public health ethics in computational modelling of infectious disease interventions. *Interface Focus*, 15, 2025.
17. Yushuf Sharker, Zaynab Diallo, **Wasiur R. KhudaBukhsh**, and Eben Kenah. Pairwise accelerated failure time regression models for infectious disease transmission in close contact groups with external sources of infection. *Statistics in Medicine*, 2024.
18. **Wasiur R. KhudaBukhsh** and Grzegorz A. Rempała. How to *correctly* fit an SIR model to data from an SEIR model? *Mathematical Biosciences*, 2024.
19. Matthew Wascher, Patrick Schnell, **Wasiur R. KhudaBukhsh**, Mikkel B.M. Quam, Joseph Tien, and Grzegorz Rempała. Estimating disease transmission in a closed population under repeated testing. *Journal of the Royal Statistical Society: Series C (JRSSC)*, 2024.
20. István Z. Kiss, Luc Berthouze, and **Wasiur R. KhudaBukhsh**. Towards inferring network properties from epidemic data. *Bulletin of Mathematical Biology*, 2024.
21. **Wasiur R. KhudaBukhsh**, Sat Kartar Khalsa, Eben Kenah, Grzegorz Rempała, and Joseph Tien. COVID-19 dynamics in an Ohio prison. *Frontiers in Public Health*, 2023.
22. **Wasiur R. KhudaBukhsh**, Caleb Deen Bastian, Matthew Wascher, Colin Klaus, Saumya Yashmohini Sahai, Mark H. Weir, Eben Kenah, Elisabeth Root, Joseph H. Tien, and Grzegorz A. Rempała. Projecting COVID-19 cases and hospital burden in Ohio. *Journal of Theoretical Biology*, 561:111404, 2023.
23. Colin Klaus, Matthew Wascher, **Wasiur R. KhudaBukhsh**, and Grzegorz Rempała. Likelihood-free dynamical survival analysis applied to the COVID-19 epidemic in Ohio. *Mathematical Biosciences and Engineering*, 20, 2023.
24. Kai Cui, **Wasiur R. KhudaBukhsh**, and Heinz Koeppl. Hypergraphon mean-field games. *Chaos*, 2022.
25. **Wasiur R. KhudaBukhsh**, Casper Woroszylo, Grzegorz Rempała, and Heinz Koeppl. A functional central limit theorem for SI processes on configuration model graphs. *Advances in Applied Probability*, 2022.
26. Colin Klaus, Matthew Wascher, **Wasiur R. KhudaBukhsh**, Joseph H. Tien, Grzegorz A. Rempała, and Eben Kenah. Assortative mixing among vaccination groups and biased estimation of reproduction numbers. *The Lancet Infectious Diseases*, 22:P579–581, 2022.
27. Francesco Di Lauro*, **Wasiur R. KhudaBukhsh***, István Z. Kiss, Eben Kenah, Max Jensen, and Grzegorz Rempała. Dynamic survival analysis for non-Markovian epidemic models. *Journal of the Royal Society Interface*, 2022. *Joint first authors.
28. Kai Cui, **Wasiur R. KhudaBukhsh**, and Heinz Koeppl. Motif-based mean-field approximation of interacting particles on clustered networks. *Physical Review E*, 105, 2022.
29. Harley Vossler, Pierre Akilimali, Yuhan Pan, **Wasiur R. KhudaBukhsh**, Eben Kenah, and Grzegorz A. Rempała. Analysis of individual-level epidemic data: study of 2018–2020 Ebola outbreak in Democratic Republic of the Congo. *Scientific Reports*, 12, 2022.
30. Ido Somekh*, **Wasiur R. KhudaBukhsh***, Elisabeth Dowling Root*, Greg Rempala, Eric Simoes, and Eli Somekh. Quantifying the population-level effect of COVID-19 mass vaccination campaign in Israel: a modeling study. *Open Forum Infectious Diseases*, 2022. *Joint first authors.
31. **Wasiur R. KhudaBukhsh**, Hye-Won Kang, Eben Kenah, and Grzegorz Rempała. Incorporating age and delay into models for biophysical systems. *Physical Biology*, 18(1), 2021. Invited paper.

32. **Wasiur R. KhudaBukhsh**, Boseung Choi, Eben Kenah, and Grzegorz Rempała. Survival dynamical systems: individual-level survival analysis from population-level epidemic models. *Journal of the Royal Society Interface Focus*, 10(1), 2020.
33. **Wasiur R. KhudaBukhsh**, Arnab Auddy, Yann Disser, and Heinz Koeppl. Approximate lumpability for Markovian agent-based models using local symmetries. *Journal of Applied Probability*, 56, 2019.
34. Hye-Won Kang*, **Wasiur R. KhudaBukhsh***, Heinz Koeppl, and Grzegorz Rempała. Quasi-steady-state approximations derived from a stochastic enzyme kinetics. *Bulletin of Mathematical Biology*, 81(5):1303–1336, 2019. *Joint first authors.
35. Saumya Yashmohini Sahai, Saket Gurukar, **Wasiur R. KhudaBukhsh**, Srinivasan Parthasarathy, and Grzegorz A. Rempała. A machine learning model for nowcasting epidemic incidence. *Mathematical Biosciences*, 2021.
36. **Wasiur R. KhudaBukhsh**, Sounak Kar, Bastian Alt, Amr Rizk, and Heinz Koeppl. Generalized cost-based job scheduling in very large heterogeneous cluster systems. *IEEE Transactions on Parallel and Distributed Systems*, 31(11):2594–2604, 2020.
37. Boseung Choi, Sydney Busch, Dieudonné Kazadi, Benoit Ilunga, Emile Okitolonda, Yi Dai, Robert Lumpkin, Omar Saucedo, **Wasiur R. KhudaBukhsh**, Joseph Tien, Marcel Yotebieng, Eben Kenah, and Grzegorz A. Rempała. Modeling outbreak data: analysis of a 2012 Ebola virus disease epidemic in DRC. *BIOMATH*, 8(2), 2019.
38. **Wasiur R. KhudaBukhsh**, Amr Rizk, Sounak Kar, and Heinz Koeppl. Provisioning and performance evaluation of parallel systems with output synchronization. *ACM Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS)*, 4(1), 2019.
39. Bastian Alt, Markus Weckesser, Christian Becker, Matthias Hollick, Sounak Kar, Anja Klein, Robin Klose, Roland Kluge, Heinz Koeppl, Boris Koldehofe, **Wasiur R. KhudaBukhsh**, Mahdi Mousavi, Martin Pfannemueller, Amr Rizk, Andy Schuerr, and Ralf Steinmetz. Transitions: a protocol-independent view of the future internet. *Proceedings of the IEEE*, 107(4):835–846, 2019.

Book chapters

40. Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, and Grzegorz A. Rempała. Functional law of large numbers for an epidemic model with random effects. Handbook of Statistics. Elsevier, 2024.
41. Grzegorz A. Rempała and **Wasiur R. KhudaBukhsh**. Dynamical survival analysis for epidemic modeling. In *Handbook of Visual, Experimental and Computational Mathematics*, pages 1–17. Springer International Publishing, 2023.

Peer-reviewed conference proceedings

42. Karim Elsayed, **Wasiur R. KhudaBukhsh**, and Amr Rizk. On the trade-off between fidelity and latency for the quantum link layer with few memories and entanglement purification. In *Proceedings of the International Conference on Quantum Communications, Networking, and Computing (QCNC 2024)*, 2024. Best Paper Award.
43. Karim Elsayed, **Wasiur R. KhudaBukhsh**, and Amr Rizk. On the fidelity distribution of link-level entanglements under purification. In *Proceedings of the IEEE International Conference on Communication (ICC) 2024*, 2024.
44. Riccardo Corradin, Luca Danese, **Wasiur R. KhudaBukhsh**, and Andrea Ongaro. Model-based clustering of non-stationary time series with common historical change times. In *Statistical Learning, Sustainability and Impact*, 2023.
45. **Wasiur R. KhudaBukhsh**, Bastian Alt, Sounak Kar, Amr Rizk, and Heinz Koeppl. Collaborative uploading in heterogeneous networks: optimal and adaptive strategies. In *IEEE International Conference on Computer Communications (INFOCOM)*, 2018. $<$20% acceptance rate. Best-in-Session Presentation Award.
46. **Wasiur R. KhudaBukhsh**, Amr Rizk, Alexander Frömmgen, and Heinz Koeppl. Optimizing stochastic scheduling in fork-join queueing models: bounds and applications. In *IEEE International Conference on Computer Communications (INFOCOM)*, 2017. $\sim$20% acceptance rate.
47. Adrian Šošić, **Wasiur R. KhudaBukhsh**, A. M. Zourbir, and Heinz Koeppl. Inverse reinforcement learning in swarm systems. In *AAMAS Workshop on Transfer in Reinforcement Learning*, 2017. Available: http://www.tirl.info/proceedings/2017/SosicEtal-Inverse.pdf.
48. Adrian Šošić, **Wasiur R. KhudaBukhsh**, A. M. Zourbir, and Heinz Koeppl. Inverse reinforcement learning in swarm systems. In *International Conference on Autonomous Agents & Multiagent Systems (AAMAS)*, 2017. $\sim$26% acceptance rate. Best Paper Award finalist.
49. **Wasiur R. KhudaBukhsh**, Julius Rückert, Julian Wulfheide, David Hausheer, and Heinz Koeppl. Analysing and leveraging client heterogeneity in swarming-based live streaming. In *IFIP Networking Conference (IFIP Networking) and Workshops*, 2016. $\sim$26% acceptance rate.
50. Mahdi Mousavi, Hussein Al Shatri, **Wasiur R. KhudaBukhsh**, Heinz Koeppl, and Anja Klein. Cross-layer QoE-based incentive mechanism for video streaming in multi-hop wireless networks. In *IEEE 86th Vehicular Technology Conference (VTC)*, 2017.

Thesis and technical notes

51. **Wasiur R. KhudaBukhsh**. *Model reductions for queueing and agent-based systems with applications in communication networks*. PhD thesis, Technische Universität, Darmstadt, 2018. Available at: http://tuprints.ulb.tu-darmstadt.de/7588/.
52. Mark Sinzger-D'Angelo, Heinz Koeppl, and **Wasiur R. KhudaBukhsh**. Bounds on the spectral radius of real-valued non-negative kernels on measurable spaces. 2023. Preprint: https://arxiv.org/abs/1808.00258.